RESEARCH GROUP R26-IT-18

Accessible Emotion-Aware E-Learning Platform

for Deaf Students Studying O/L Information Technology

Using Real-Time Attention Monitoring, Knowledge Graph-Based Adaptive Learning, an Emotion-Aware Chatbot, Sign Avatar Support and Smart Haptic Feedback



THE PROBLEM

Traditional e-learning leaves deaf learners unseen

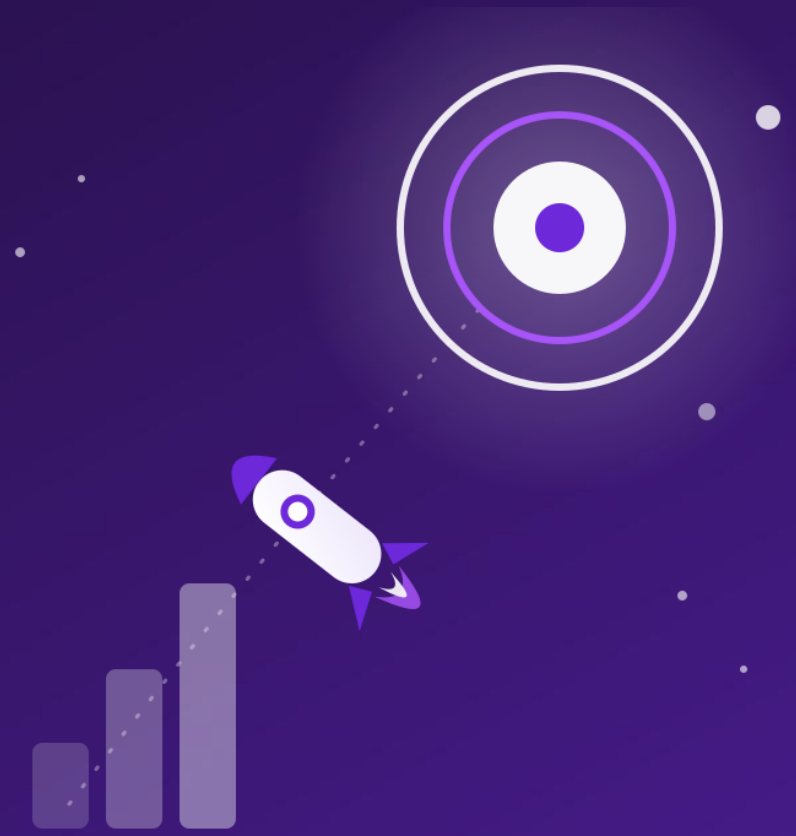
Traditional e-learning platforms are mainly designed for hearing students and commonly depend on audio explanations, spoken instructions and text-heavy learning materials. These platforms are unable to identify when a deaf learner becomes distracted, confused, drowsy or misses an important learning concept.

As a result, attention loss, missed concepts and non-audio needs go unaddressed by existing tools.

- Student attention loss may remain unnoticed.
- Missed lesson sections are not automatically identified.
- Learning support is usually static and not personalized.
- Generic quizzes are not connected to the exact concepts a student missed.
- Technical ICT concepts may be difficult to understand using text alone.
- Traditional audio alerts are unsuitable for deaf learners.
- Teachers have limited visibility into individual learning difficulties.

MAIN OBJECTIVE

To develop an accessible and intelligent e-learning platform for deaf and hard-of-hearing O/L ICT students that monitors learner attention in real time, identifies missed or difficult learning concepts, and provides personalized learning support through adaptive questions, an emotion-aware chatbot, sign-language visualization, and smart haptic feedback.



SUB OBJECTIVES

Five pillars of the platform

01

Real-Time Attention Monitoring and Telemetry

Monitor student focus and detect distraction or missed lesson periods in real time.

02

Knowledge Graph-Driven Adaptive Question System

Generate relevant popup questions based on the current ICT concept, prerequisites, and related topics.

03

Emotion-Aware Adaptive Chatbot

Provide personalized explanations, revision support, and assessments based on the student's learning state.

04

Sign Avatar Lecture Generator

Convert important ICT lesson content into animated sign-language visual support.

05

Smart Haptic Wristband and Sign Practice

Provide vibration-based learning alerts and support interactive camera-based sign practice.

OVERALL PROJECT

What the platform delivers

- Build an accessible AI-powered learning platform for deaf O/L ICT students.
- Monitor attention and detect missed lesson segments in real time.
- Identify weak ICT concepts and provide adaptive learning support.
- Deliver personalized chatbot, quiz, and revision assistance.
- Improve accessibility through sign-language visuals and haptic feedback.
- Track attention, performance, and concept mastery.
- Integrate all features into one adaptive learning system.

STAKEHOLDERS

Primary Users

Deaf and hard-of-hearing O/L ICT students

Secondary Users

ICT teachers, special education schools, inclusive education institutions, parents and guardians.

RESEARCH GAP

What existing systems still miss

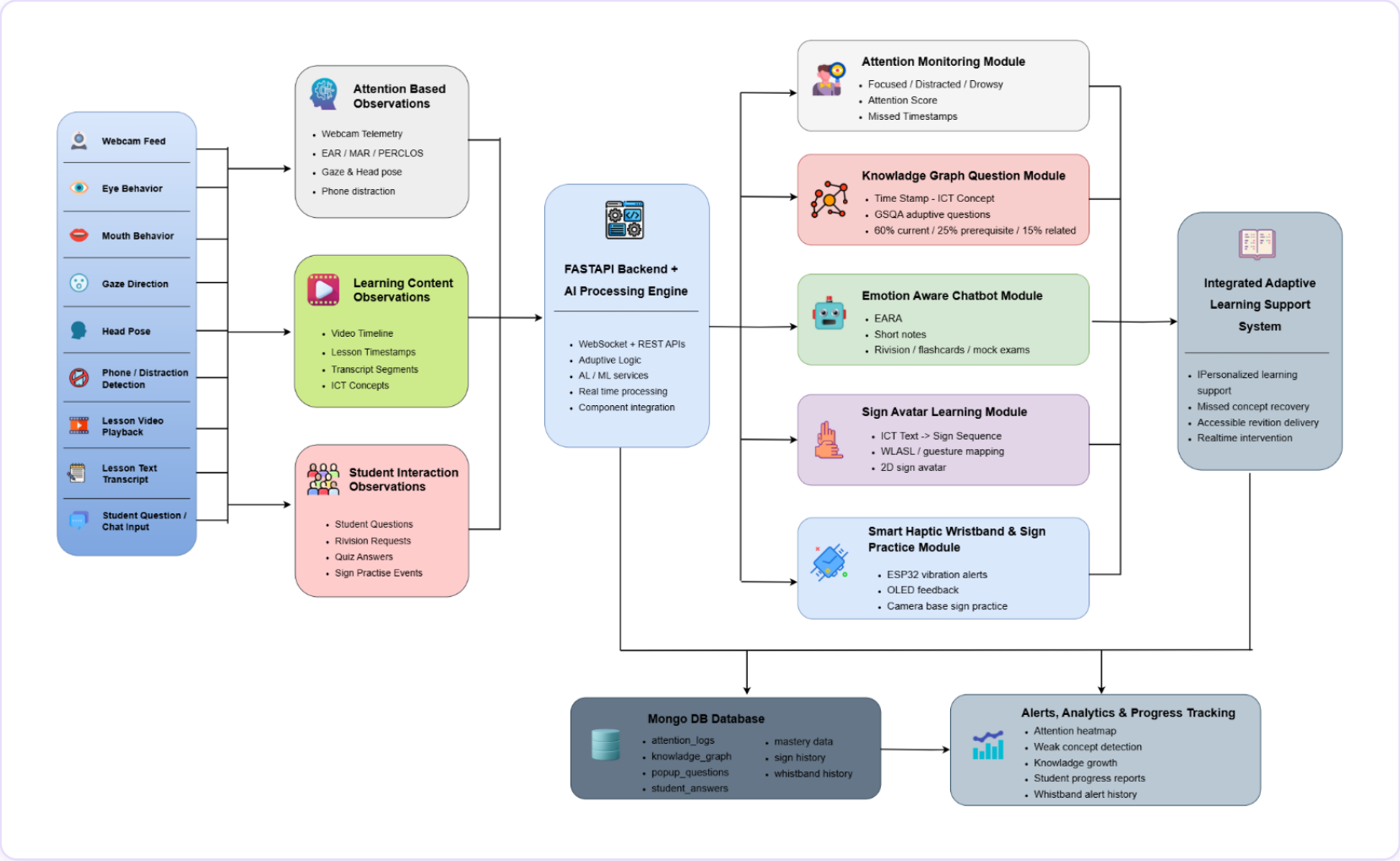
Most existing e-learning systems are still designed as static learning platforms and do not fully support the needs of deaf students. They generally cannot detect when a student loses attention, adapt learning content according to the student's learning state, or provide continuous sign-language-based learning support.

Existing systems often:

- Cannot detect attention, distraction, or drowsiness in real time.
- Cannot identify the exact missed lesson concept.
- Do not adapt learning based on the student's learning state.
- Use generic chatbots with limited personalization.
- Provide limited sign-language support for ICT content.
- Use static or random quizzes instead of adaptive questions.
- Lack non-audio feedback for deaf learners.
- Do not integrate attention, learning performance, revision, and accessibility in one system.

SYSTEM ARCHITECTURE

System Overview & Integration



DEVELOPMENT STATUS

Integration & Completion Status

Real-Time Attention Monitoring & Telemetry

Implemented

- Webcam-based attention monitoring
- EAR and MAR calculation
- PERCLOS drowsiness detection
- Gaze and head-pose monitoring
- YOLOv8 phone/distraction detection
- Attention score and timeline heatmap
- MongoDB attention logging

Emotion-Aware Adaptive Chatbot

Implemented

- EARA response adaptation
- Attention-based weak-spot recommendations
- Multiple explanation styles | O/L short notes
- Knowledge growth tracking
- Past-paper grader
- SM-2 flashcards
- Mock examination

Sign Avatar Lecture Generator

Implemented

- 30 ICT sign vocabulary items
- Landmark-driven 2D avatar
- WLASL BiLSTM integration
- Sign MNIST pipeline
- Gesture fallback mechanism

Smart Wristband & Sign Practice

Implemented

- ESP32 haptic wristband
- Multiple vibration patterns
- OLED feedback
- Virtual wristband simulation
- Camera-based sign practice
- Course progression tracking

COMMERCIALIZATION

Who it's for & the value it delivers

CURRENT TARGET USERS

- Deaf and hard-of-hearing O/L ICT students
- Special education schools
- Inclusive education institutions
- ICT teachers
- Educational organizations
- Government inclusive-education programmes
- NGOs supporting deaf education

Value Proposition

A unified accessible learning platform that does not simply deliver educational content, but detects when learning is interrupted and automatically provides personalized learning recovery.

- Real-time attention awareness
- Personalized concept revision
- Curriculum-aware AI assistance
- Sign-language learning support
- Non-audio haptic intervention
- Student progress and learning analytics

COMMERCIALIZATION

Sustainability & Revenue Model

Monthly or annual school subscription

Cloud-based SaaS educational platform

Institutional licensing

Optional ESP32 smart wristband package

Government and NGO partnerships

Premium student and teacher analytics

These commercialization approaches are also identified in the original project proposal.

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Thank You

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